Clinical Data Interchange Standards Consortium
CDISC) Standards in SDTMs and ADaMs

Lesson 8: Creating and QCing DEFINE. XML

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SAS's Circle of Excellence Award

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ODM: The Big Picture

Different Purpose

1) Documentation **2) Attributes** **3) Clinical Data**

DEFINE.PDF

SDTM Variables

ADaM Variables

Control Terms

Value-Level

DEFINE.XML

PLUS

Metadata

Replaces DEFINE.PDF, Style sheet also required

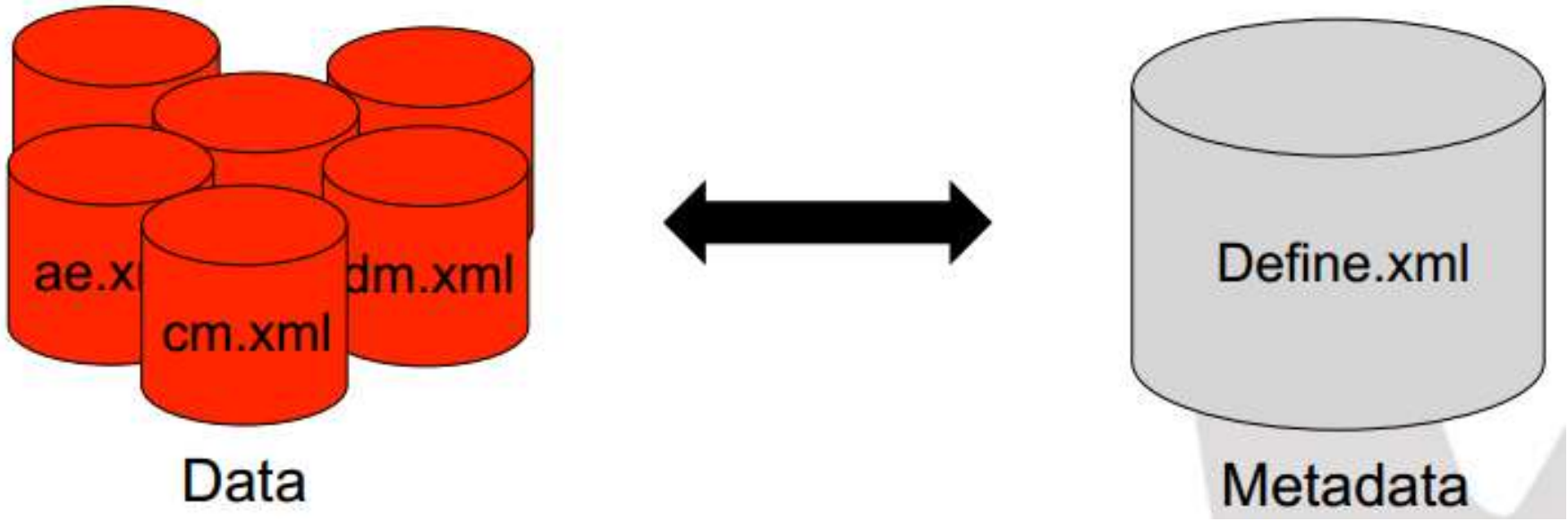
Dataset-XML File

SDTM and ADaM Metadata and Data

Dataset-XML may replace SAS transport files

Features of Dataset-XML and Define.xml

- ✓ Dataset-XML is an extension of ODM
- ✓ One XML-file per dataset
- ✓ Metadata stored outside the dataset (Define.xml)



Currently still transport files

Creating and QCing Define.xml & Dataset-XML

Source	Tools	Target	QC Methods
Excel files, SAS datasets storing metadata	Formedix, OpenCDISC, DATA Step (DATA _NULL_, FILE / PUT)	**DEFINE.XML**	- SAS XML mapper to view XML files and create datasets - XMLpad to view XML files - Formedix - OpenCDISC
SDTMs/ ADaMs	FILENAME / LIBNAME XML, PROC CDISC, SAS Clinical Standards Toolkit	**DATASET-XML files ex. DM.XML**	- FILENAME / LIBNAME XML to convert XML back to dataset - One dataset-xml file per SDTM/ADaM

QC Checks include both ODM schema tags and edit checks.
ODS and Proc Template can be used to create custom tags.

Creating DEFINE.XML from SAS

Convert **Create**

Excel* files, SAS Datasets

- Defined by schema (.xsd)

OpenCDISC, DATA Step

- DATA Step to create tags, based on schema and assign values (admin)*

- PROC Contents for attributes and format catalog

DEFINE.XML

- SAS XML Mapper to view XML file and create SAS datasets

- XMLpad to view XML file

* Excel may be used to store some DEFINE.XML metadata. ODS and Proc Template can be used to create custom tags.

Creating Dataset.XML from SAS

1) SAS Datasets	2) Three Methods	3) Export
SDTMs/ ADaMs	**a) FILENAME/ LIBNAME XML** (Import/Export) **b) PROC CDISC** (Export) **c) SAS Clinical Standards Toolkit**	**Dataset.XML, ex. DM.XML** **a) Direct** **b) With ODM tag values**
- Patient data values	- Importing XML files to create SAS datasets are useful to qc XML files	- One Dataset.xml file per SDTM/ADaM - Dataset.XML files may replace SAS transport files

Anatomy of Dataset.xml File

Dataset.XML File

Level

1 SDTM Attributes

2

USUBJID 1 data values	USUBJID 2 data values	USUBJID 3 data values

One Dataset.XML per SDTM and ADaM.

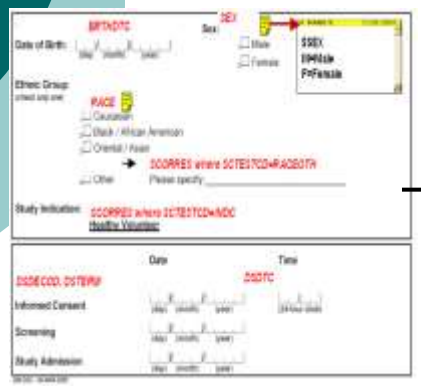
Interchange between Dataset.XML and SAS Datasets (Direct **without** additional ODM tags)

Task	SAS Syntax
Create **Dataset.xml** file from SAS dataset	libname MYSAS 'c:\sas datasets\'; **libname MYXML XML 'c:\xml datasets\dm.xml';** **data MYXML.DM**; **set MYSAS.DM**; run;
Create **Dataset.xml** file from SAS dataset	**ods markup body='c:\my documents\xml\class.xml';** proc print data=**sashelp.class**; run; ods markup close;
Create **SAS Dataset** from Dataset.xml file for qc purpose	libname MYSAS 'c:\sas datasets\'; **libname MYXML XML 'c:\xml datasets\dm.xml';** **data MYSAS.DM**; **set MYXML.DM**; run;
Create **SAS Dataset** from ODM-xml file with ODM tags for qc purpose	**filename ODM 'c:\myproject\xml\AE.XML';** **libname ODM XML XMLTYPE=CDISCODM** **formatActive=YES formatNoReplace=NO** **formatLibrary="Work";** **data AE; set ODM.AE**; run;

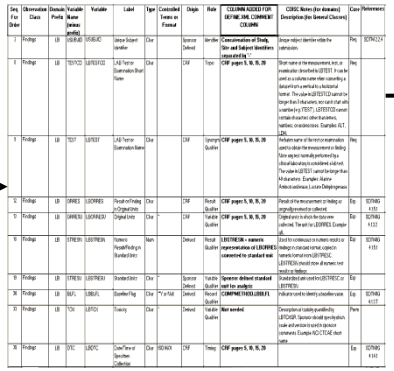
Use SAS XML Mapper for converting between complex XML files and SAS datasets. See SAS paper on XML engine.

SDTM Process Flow

Annotated CRF

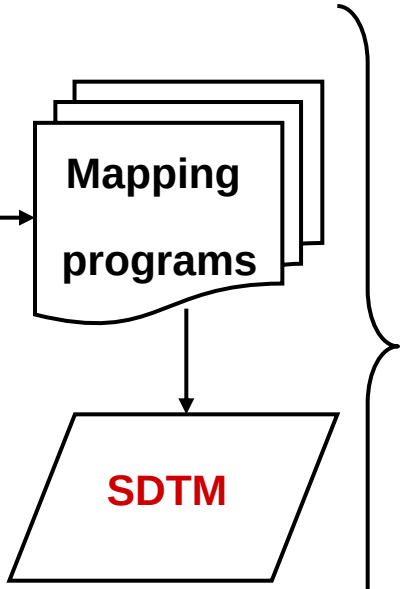


SDTM Specifications



(dataset templates)

(source for define.xml)



define.xml (SDTM)

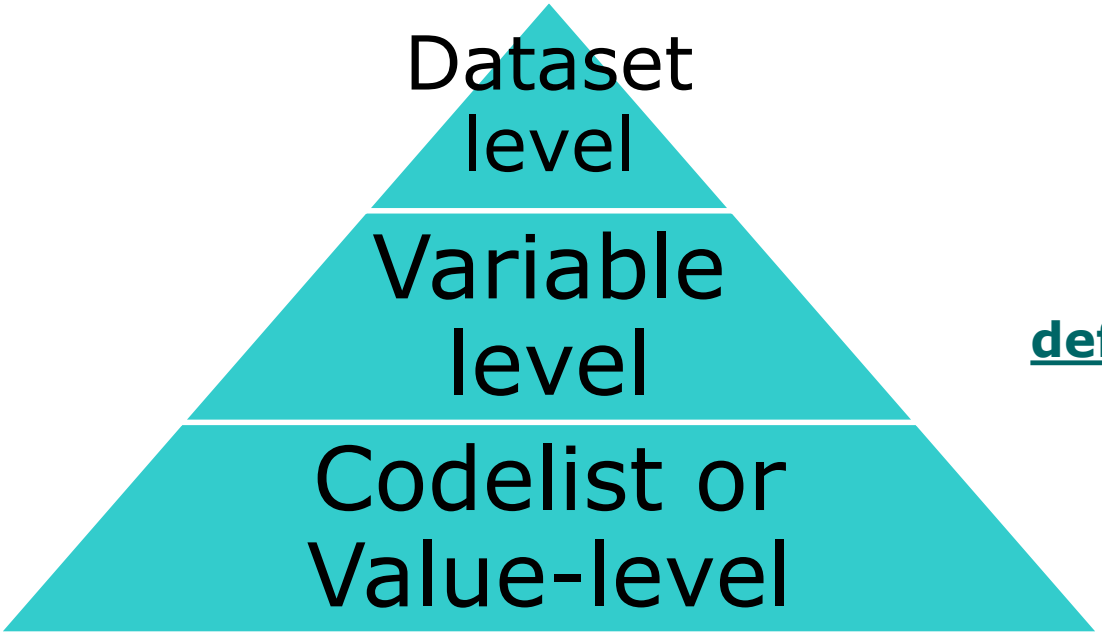
red: e-submission deliverables

Communication

SDTM DEFINE.PDF – Dataset Level

Datasets for Study CDISC01						
Dataset	**Description**	**Class**	**Structure**	**Purpose**	**Keys**	**Location**
TA	Trial Arms	Trial Design	One record per planned Element per Arm	Tabulation	STUDYID, ARMCD, TAETORD	ta.xpt
TE	Trial Elements	Trial Design	One record per planned Element	Tabulation	STUDYID, ETCD	te.xpt
TI	Trial Inclusion/Exclusion Criteria	Trial Design	One record per I/E criterion	Tabulation	STUDYID, IETESTCD	ti.xpt
TS	Trial Summary	Trial Design	One record per trial summary parameter value	Tabulation	STUDYID, TSPARMCD, TSSEQ	ts.xpt
TV	Trial Visits	Trial Design	One record per planned Visit per Arm	Tabulation	STUDYID, VISITNUM, ARMCD	tv.xpt
DM	Demographics	Special Purpose	One record per subject	Tabulation	STUDYID, USUBJID	dm.xpt
SE	Subject Elements	Special Purpose	One record per actual Element per subject	Tabulation	STUDYID, USUBJID, SESTDTC, SEENDTC, ETCD	se.xpt

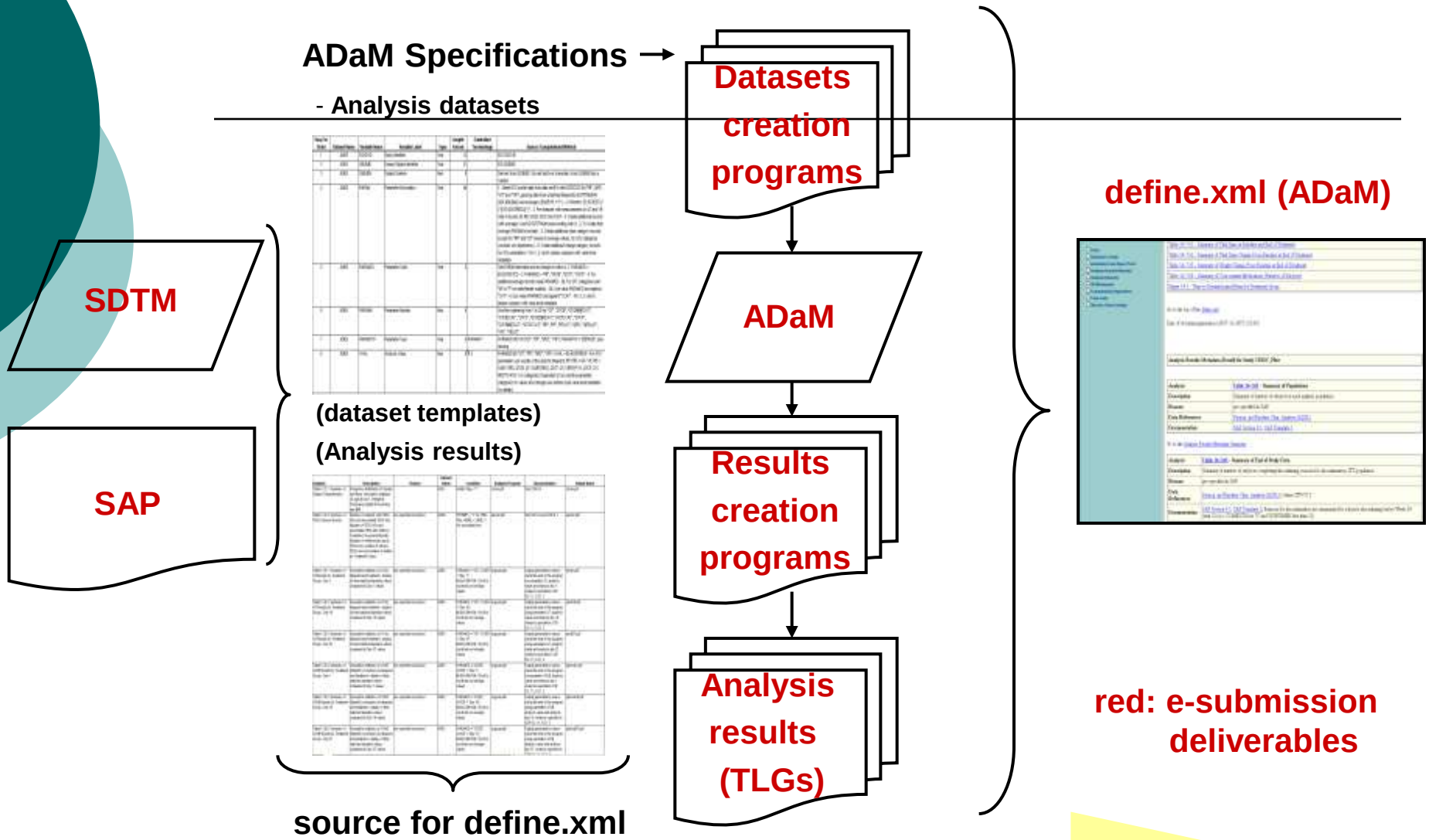
Order of datasets is important



See define.xml

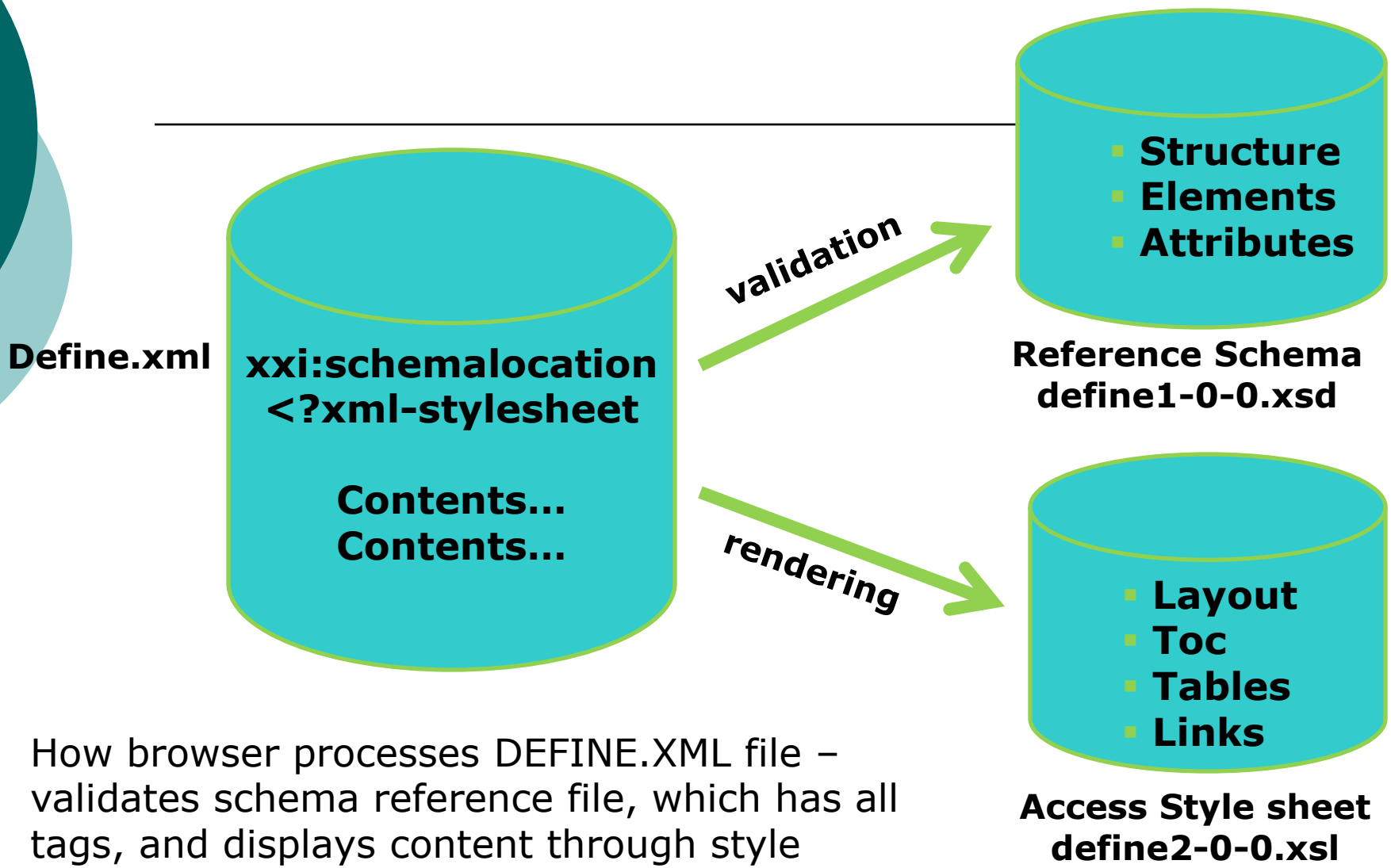
Hyperlinks for more information.
Required documentation to be delivered with FDA submission.

ADaM Process Flow



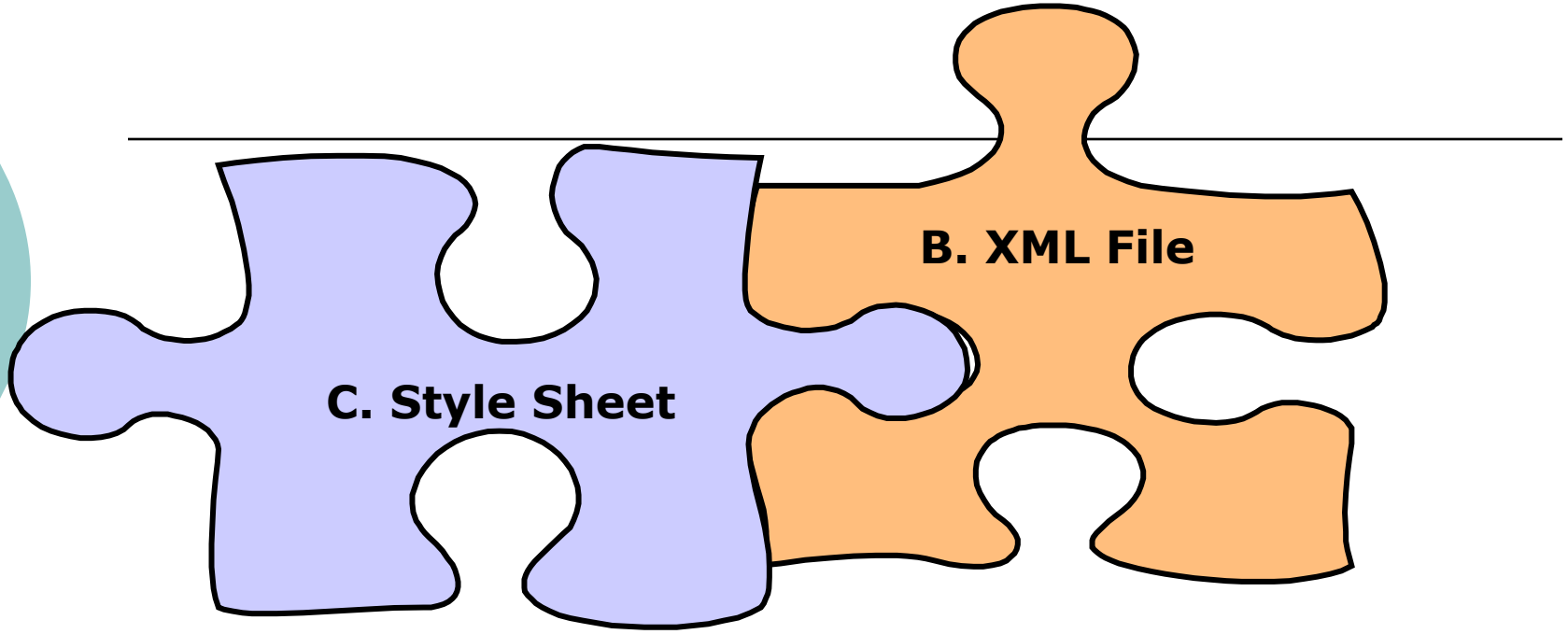
Communication

DEFINE.XML File Components



How browser processes DEFINE.XML file – validates schema reference file, which has all tags, and displays content through style sheet, which has color, fonts, etc.

XML Basics – Three reference files



(define2-0-0.xsl)

Controls structure display of each
A. Schema tags
(define1-0-0.xsd)

(define.xml)

Contains metadata content for each
A. Schema tag
(define1-0-0.xsd)

All three files reference each other and must be consistent with each other.

XML Basics – Three reference files

- **A. Schema (define1-0-0.xsd referenced on cdisc.org)**
  - Controls content, rules for names, attributes, order, etc.
  - Declaration of valid elements (tags) and their attributes
  - Prerequisite for machine-readability
  - Similar to Document Type Definition (DTD)

- **B. XML File (define.xml created by SAS program)**
  - Data and metadata in machine-readable format
  - Usage of elements and attributes as defined in schema

- **C. Style Sheet (define2-0-0.xsl from cdisc.org)**
  - Controls how XML file is displayed
  - Definition of layout in browser tool for Human-Readability
  - Usage of elements and attributes as defined in schema

A. Schema (define1-0-0.xsd)

A schema for a define.xml file defines the:

1. The elements that can appear
2. The attributes that can appear for elements
3. Which elements are child elements
4. The order (structure) of child elements
5. The number of child elements
6. Whether an element is empty or can include text
7. The data types for elements and attributes
8. Default and fixed values for elements and attributes

Check if Define.xml Does Not Conform to CDISC Define.xml Schema • Required XML field (such as the domain level 'Repeating' field or the variable level 'DataType' field) is missing (High severity) • Value of required field is not valid (For example, valid values for the field DataType are: text, integer, float, date, time, or datetime) (High severity)

A. Schema (define1-0-0.xsd)

XML schemas are based on ODM and CDISC standards, but they are extensible.

<u>Using a schema makes it easy to:</u>

1. Describe allowable file content
2. Validate the correctness of data
3. Work with data from databases
4. Define data aspects (restrictions on data)
5. Define data patterns (data formats)
6. Convert data to different data types

A. Schema (**define1-0-0.xsd)**

define_extension.xsd - Notepad

File Edit Format View Help

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<xs:schema targetNamespace="http://www.cdisc.org/ns/odm/v1.2"
           xmlns="http://www.cdisc.org/ns/odm/v1.2"
           xmlns:xs="http://www.w3.org/2001/XMLSchema"
           xmlns:ds="http://www.w3.org/2000/09/xmldsig#"


           xmlns:def="http://www.cdisc.org/ns/def/v1.0"
           xmlns:odm="http://www.cdisc.org/ns/odm/v1.2"
           elementFormDefault="qualified" attributeFormDefault="unqualified">


    <xs:import namespace="http://www.cdisc.org/ns/def/v1.0" schemaLocation="http://www.cdisc.org/schema/def/v1.0/define-ns.xsd"/>


    <xs:redefine schemaLocation="http://www.cdisc.org/schema/odm/v1.2.1/ODM1-2-1.xsd">
```

In general, schemas are used to validate:

- **Data completeness.** Schema validation checks to ensure all information required by the schema is present in the *define.xml* instance.
- **Data structure.** Schema validation checks to ensure that the basic structure of the elements and attributes in the *define.xml* instance matches the schema.
- **Data correctness.** Schema validation indicates that the data conforms to the rules of the schema. However, it does not guarantee that the information is meaningful from the consumer's perspective.

Access schema from cdisc.org, See XML Schema Validation for Define.xml

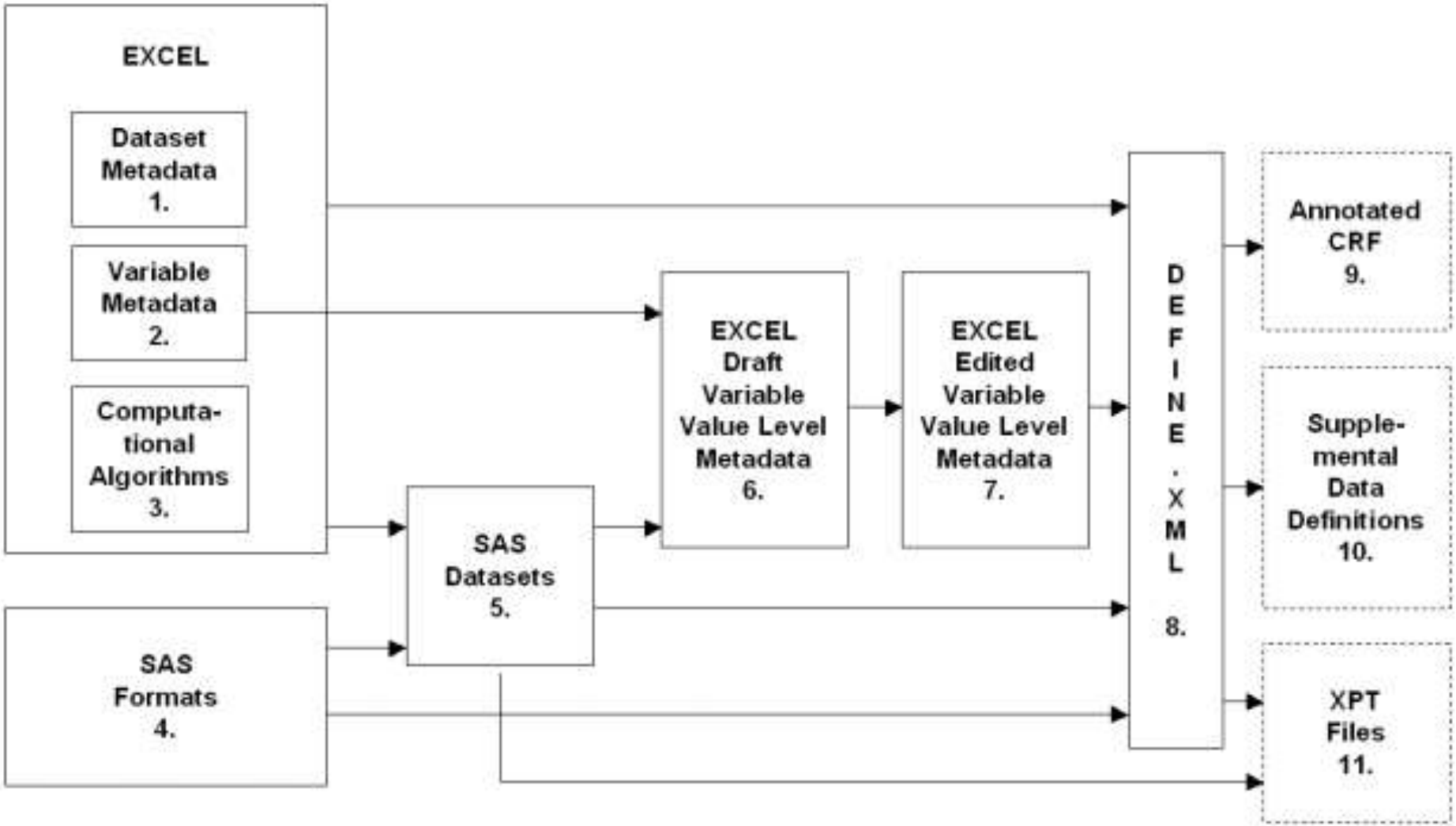
B. DEFINE.XML Process Flow with Excel

A	B	E	F
Parameter_Description	**Parameter_Value**		
xmlVersion	1.0		
xmlLanguage	ISO-8859-1		
StyleSheetType	text/xsl		
StyleSheetPath	UTIL/XSL/cp01.xsl		
CDISC_ODM_ReferenceDocument	http://www.cdisc.org/ns/odm/v1.3		
W3_XML_SchemaDocument	http://www.w3.org/2001/XMLSchema-instance		
W3_ReferenceDocument	http://www.w3.org/1999/xlink		
CDISC_CRT_SchemaExtensions	http://www.cdisc.org/ns/crt/v3.1.1		
CDISC_SDTM_ModelReference	http://www.cdisc.org/ns/sdtm/v3.1.1		
CDISC_Pilot_CP01_Namespace	http://www.cdisc.org/ns/pilot/1.0		
CDISC_DEF_DocumentReference	http://www.cdisc.org/ns/def/v1.0		
SchemaLocation	http://www.cdisc.org/ns/odm/v1.3 UTIL/cp01.xsd		
ODM_Version	1.3		
FileOID	CDISC ADaM		
FileType	Snapshot		
CreationDate_time	2010-06-04T10:00:00		
AsOfDate_time	2010-05-26T02:39:00		
StudyOID	BIPI XXXX.yy		
StudyName	BIPI XXXX.yy		
StudyDescription	A phase III Randomised, double-blind, double-dummy, placebo-controlled, 4-way cross-over study to determine the 24-hour FEV1-time profiles of an orally inhaled drug		
ProtocolName	BIPI XXXX.yy		
MetaDataOID	BIPI XXXX.yy		
MetaDataVersionName	Study BIPI XXXX.yy , Data Definitions		
MetaDataDescription	Study BIPI XXXX.yy, Data Definitions		
SchemaVersionUsed	1.0.0		
CDISC_StandardName	CDISC ADaM		
CDISC_StandardVersion	2.0		

Header | Domains | Domain_Variables | Code_lists | Computation_Methods | Document_links | Results_metadata

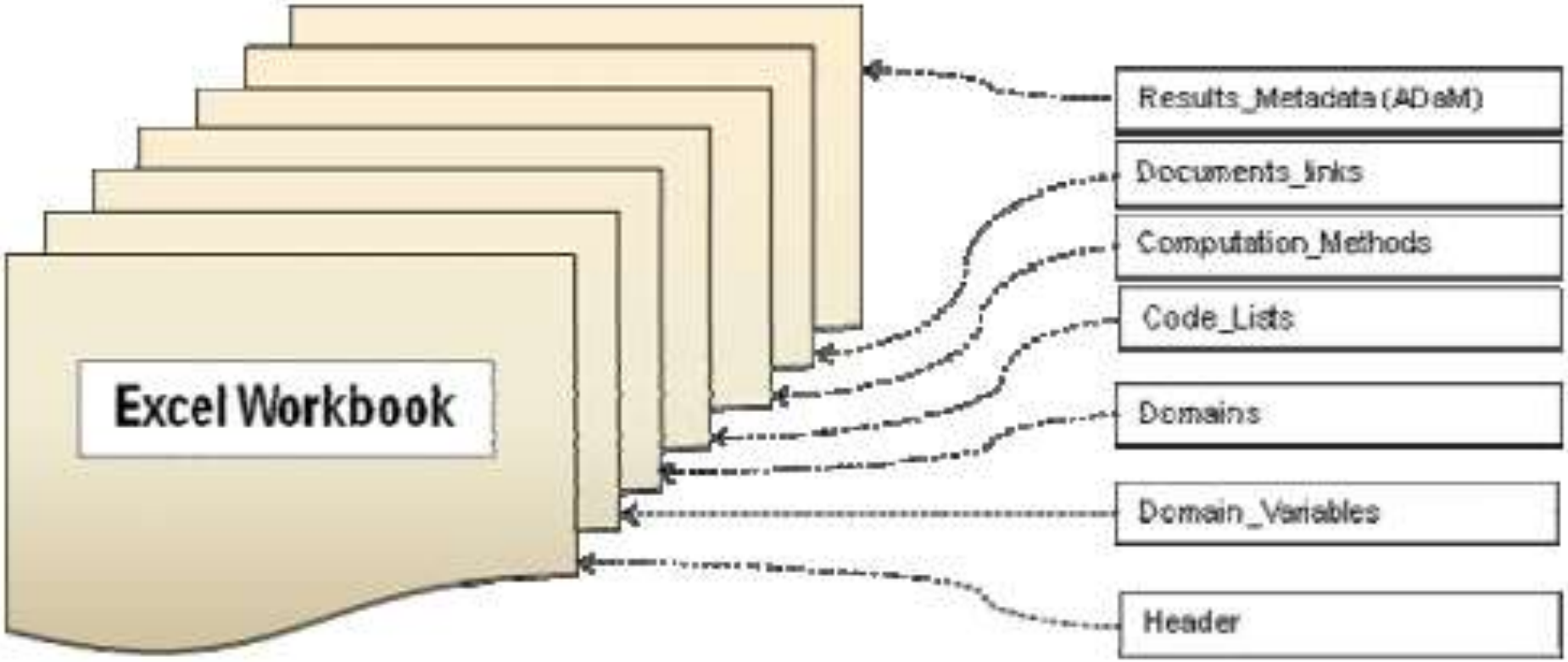
See Creating a define.xml file for ADaM and SDTM SAS Paper

B. DEFINE.XML Process Flow with Excel



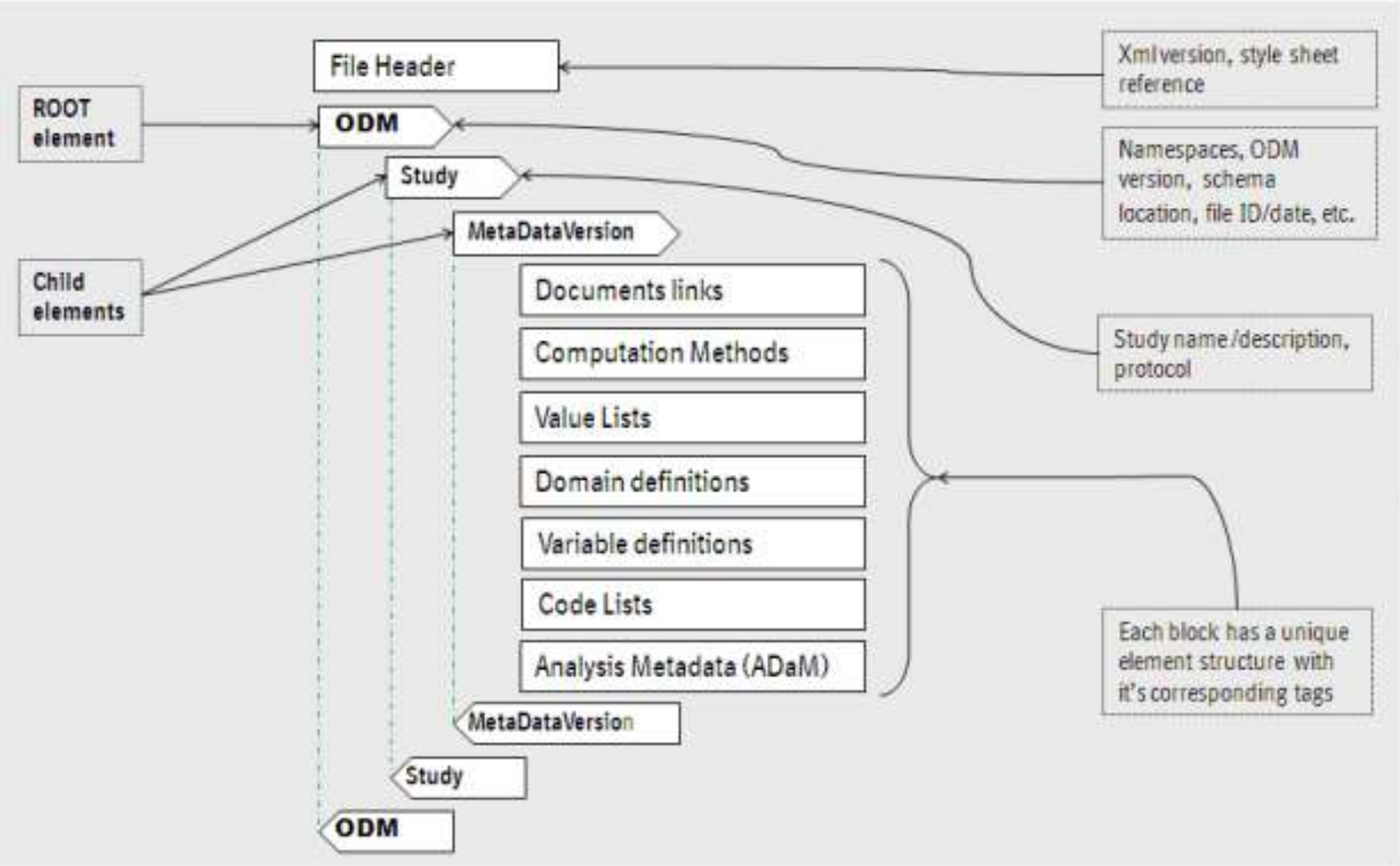
Excel file with DATA steps and PUT statements can be used to create a Define.XML file in SAS using dataset, variable and value-level metadata and codelists/method information.

B. DEFINE.XML Process Flow with Excel

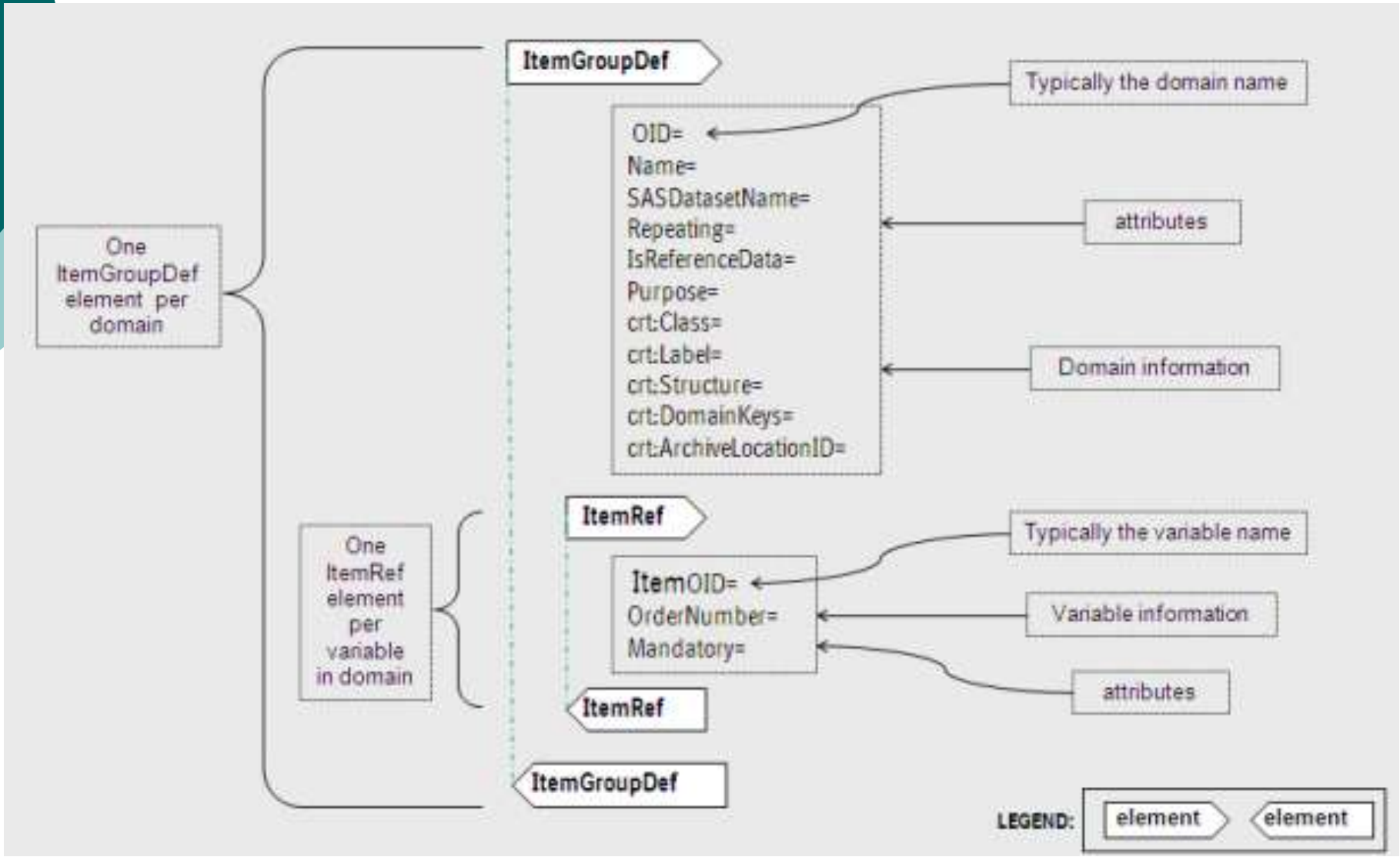


Separate sheets for each metadata.

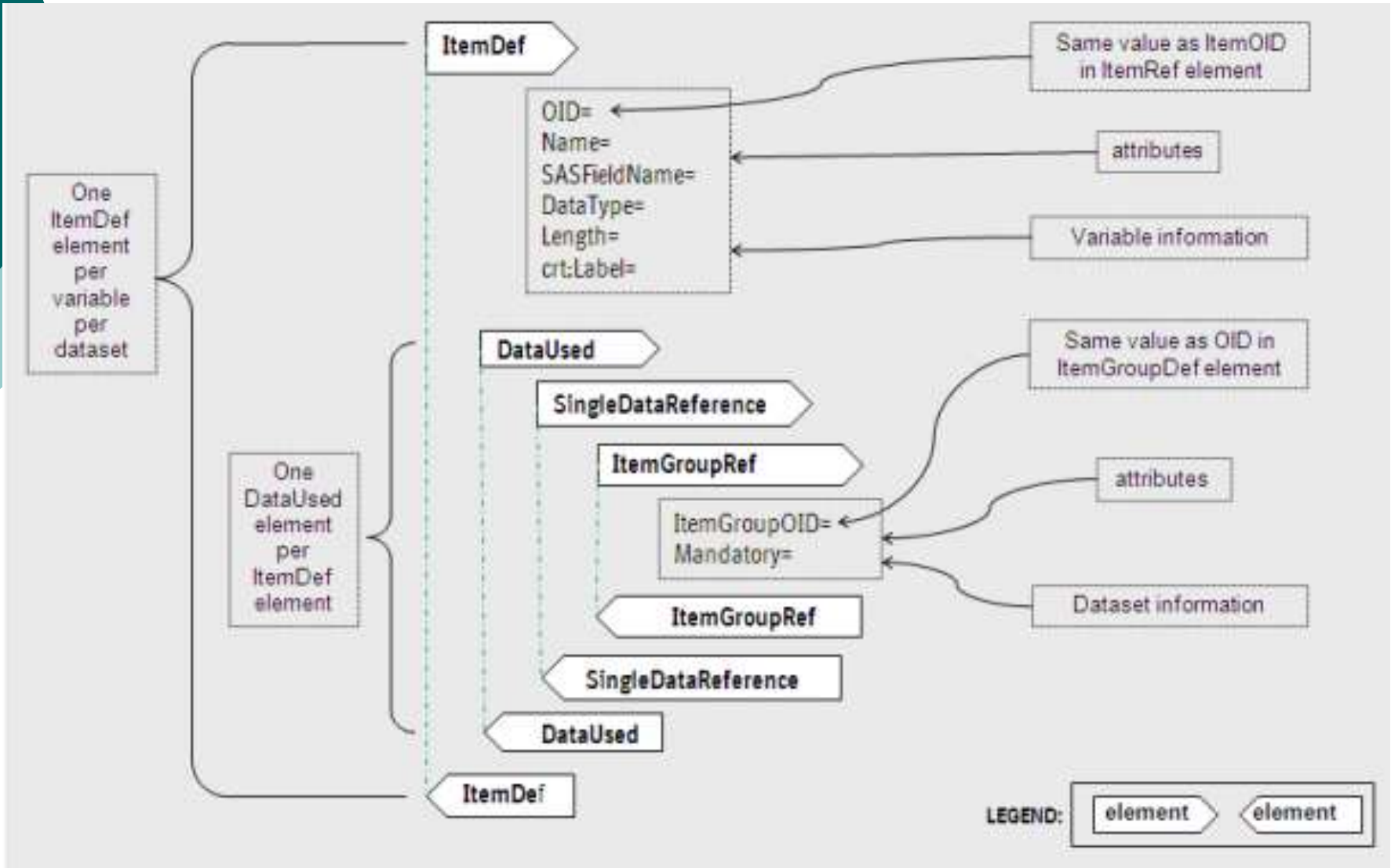
B. DEFINE.XML – ODM Element



B. DEFINE.XML – ItemGroupDef Element – One per Dataset



B. DEFINE.XML – ItemDef Element – One per Variable



Define-XML Standard

- **Metadata exchange mechanism**, based on the Operational Data Model (ODM) Standard
  - any tabular data structure
- Machine readable
  - XML tags - elements and attributes
  - Sections -building blocks- can be parsed to display different views or to process at receiving end
- Must be well formed, comply with XML schema and remaining specification
- Properly reference standards used (CDISC values - extensible)
- ***Still evolving …***

Define-XML 2.0 document (define.xml)
Example - starting and ending segments

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="define2-0-0.xsl"?>

<ODM
  xmlns="http://www.cdisc.org/ns/odm/v1.3"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns:def="http://www.cdisc.org/ns/def/v2.0"
  ODMVersion="1.3.2"
  FileOID="www.sponsor.com.Study01Test_Define-
   XML_2.0.0"
  FileType="Snapshot"
  …
  CreationDateTime="2015-10-10T21:36:09">

  <Study OID="Study01Test">
    <GlobalVariables>
      <StudyName>Study01Test</StudyName>
      <StudyDescription>A Phase #, Randomized
       …
      </StudyDescription>
      <ProtocolName>Study01Test</ProtocolName>
    </GlobalVariables>
    <MetaDataVersion
    OID="MDV.Study01Test.SDTM.1.4.SDTMIG.3.2"
      Name="Study Study01Test, Data Definitions"
      Description="Study01Test, Data Definitions"
      def:DefineVersion="2.0.0"
      def:StandardName="SDTM-IG"
      def:StandardVersion="3.2">

  <!-- ********************************************** -->
  <!-- Reference to specific CDISC Terminology        -->
  <!-- Version not in 2.0, expected in 2.1            -->
  <!-- ********************************************** -->

…

  </MetaDataVersion>
   </Study>
</ODM>
```

Define-XML 2.0 document - Variable and Value Level Metadata

Define-XML 2.0 document

```
<ItemDef OID="IT.VS.VSSTRESC" Name="VSSTRESC" DataType="text" Length="200"
SASFieldName="VSSTRESC">
 <Description>
   <TranslatedText xml:lang="en">Character Result/Finding in Std Format</TranslatedText>
 </Description>
 <def:Origin Type="Derived"/>
  <def:ValueListRef ValueListOID="VL.VS.VSSTRESC"/>
</ItemDef>
…
<ItemDef OID="IT.VS.VSSTRESC(BMI)" Name="VS.VSSTRESC - VL Item 0001"
  DataType="float"
  Length="6" SignificantDigits="2" def:DisplayFormat="6.2">
  <Description>
    <TranslatedText xml:lang="en">
      Character Result/Finding in Std Format -  WC.VS.VSSTRESC(BMI)
    </TranslatedText>
  </Description>
  <def:Origin Type="Derived"/>
</ItemDef>
```

The Viewer Software (e.g, stylesheet)

- **Must NOT transform content!**
- **Should be validated!**

Vital Signs (VS) [Location: vs.xpt]

Variable	Label	Key	Type	Length	Controlled Terms or Format	Origin
VSSTRESC	Character Result/Finding in Std Format		text	200		Derived

Display using default stylesheet

Value Level Metadata - VS [VSSTRESC]

Variable	Where	Type	Length / Display Format	Controlled Terms or Format	Origin	Derivation/Comment
VSSTRESC	VSTESTCD EQ BMI (Body Mass Index)	float	6.2		Derived	round((VSSTRESN for weight/ (VSSTRESN for height / 100)^2), .01). Note: height and weight at the closest visit collected.

SAS Program to create DEFINE.XML (Partial)

```
%MACRO escapechars (char_str=);
 &char_str = TRANWRD (&char_str, '&', '&amp;');
 &char_str = TRANWRD (&char_str, "<", '&lt;');
 &char_str = TRANWRD (&char_str, ">", '&gt;');
 &char_str = TRANWRD (&char_str, "'", '&apos;');
 &char_str = TRANWRD (&char_str, "", '&quot;');
 &char_str = TRANWRD (&char_str, '0a'x, '&#xA;');
%MEND;

libname xlsdata 'F:\CDISC\1222  Project\PharmaSug\datasets\analysis\User_defined_xml_input.xls'
          getnames=yes scantext=yes scan_textsize=yes version=2002;

data header;
 set xlsdata.'Header$'n;
run;

data Domains ;
 set xlsdata.'Domains$'n  ;
 length item_group_oid $50;
 item_group_oid=trim(dataset_name);
 if dataset_name ^= ' ';
 %escapechars(char_str=dataset_description);
run;
proc sort data=Domains; by dataset_name;  run;

data Domain  variables ;
 set xlsdata.'Domain_variables$'n;
 length item_oid item_group_oid $50;
 item_group_oid=trim(dataset_name);
 item_oid=trim(dataset_name)||'.'||trim(variable_name);
 %escapechars(char_str=variable_label);
run;
proc sort data=Domain_variables; by dataset_name variableorder;  run;

data Code_lists;
 set xlsdata.'Code_lists$'n;
 length codelist_oid $50;
 codelist_oid='Codelist.'||trim(codelist_id);
 %escapechars(char_str=decode_text);
 if codelist_id ^=' ';
run;
```

Macro to replace non-allowable characters in XML elements

Read in each spreadsheet and do some pre-processing

C. StyleSheet (**define2-0-0.xsl)**

```
▼<xsl:stylesheet xmlns:odm="http://www.cdisc.org/ns/odm/v1.2" xmlns:xsl="http://www.w3.org/1999/XSL/Transform" xmlns:xsi="http://www.w3c.org/2001/XMLSchema-instance" xmlns:def="http://www.cdisc.org/ns/def/v1.0"
 xmlns:xlink="http://www.w3c.org/1999/xlink" version="1.0">
  <div cotype="cs" id="cosymantecbfw" style="width: 0px; height: 0px; display: none;" fid="2036696279"/>
  <xsl:output method="html" version="4.0" encoding="ISO-8859-1" indent="yes"/>
 ▼<!--
    ******************************************************************************************
  -->
  <!--  File: define1-0-0.xsl                                                                  -->
  <!--  Date: 28-01-2005                                                                       -->
  <!--  Version: 1.0.0                                                                         -->
 ▼<!--
    Author: William Qubeck (Pfizer), William Friggle (Sanofi-Synthelabo), Anthony Friebel (SAS)
  -->
 ▼<!--
    Organization: Clinical Data Interchange Standards Consortium (CDISC)
  -->
 ▼<!--
    Description: This is a Style Sheet for the Case Report Tabulation Data Definition Specification
  -->
 ▼<!--
     Version 1.0.0.  This document is compliant with XSLT Version 1.0 specification (1999).
  -->
 ▼<!--
    Notes:  The define.xml document can be rendered in a format that is human readable if it contains an
```

Download stylesheet from cdisc.org

C. StyleSheet (**define2-0-0.xsl)**

A sample top-level Style sheet:

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<xsl:stylesheet version="1.0" xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
  <xsl:output method="html" version="4.0" encoding="ISO-8859-1" indent="yes"/>

  <xsl:template match="/">

    <html>
      <head>
        <script language="vbs">
          document.cookie="xmlfile="+replace(document.url,"%20"," ")
        </script>
      </head>

      <frameset cols="23%,77%" frameborder="1">
        <frame name="toc" src="UTIL/xsl/cp01_toc.htm"/>
        <frame name="contents" src="UTIL/xsl/define_contents.htm"/>
      </frameset>

    </html>

  </xsl:template>
</xsl:stylesheet>
```

Create a HTML file for display

Left frame 23% wide
Right frame 77% wide

Two frames, each defined by its own style sheet

ToC

Download stylesheet from cdisc.org

Behind the Scenes - Text file of DEFINE.XML

XML Basics, Illustrated


```
<?xml version=
- <ODM Creation                                                    CDISCPILOT
http://www.cdis                                                    s/def/v1.0"
xmlns:xsi="http://www.w3.org/2001/XM                               odm/v1.2"
  - <Study OID="CDISCPILOT01">
    - <GlobalVariables>
        <StudyName>CDISCPILOT01</StudyName>
        <StudyDescription>CDISCPILOT01 Data Definition</StudyDescription>
        <ProtocolName>CDISCPILOT
      </GlobalVariables>
    - <MetaDataVersion OID="C                                      Name="CDISC S
      Data Definitions" Name=
        - <def:AnnotatedCRF>
            <def:DocumentRef leafID="blankcrf"/>
          </def:AnnotatedCRF>
        - <def:leaf xlink:href="blankcrf.pdf" ID="blankcrf">
            <def:title>Annotated Case Report Form</def:title>
          </def:leaf>
          <def:Computati                                           ">Total scores of "ALZHEIMER'S D
            INVENTOR                                               d in the SAP. For all other Test cod
            numeric</
          <def:ComputationMeth                                     MPMETHOD.STUDY_DAY">(date portion of --DTC) minus (date p
            RFSTDC</def:C                                          onMethod
        - <def:ValueListDef                                        ="ValueList.LB.
            <ItemRef Mandatory="No" OrderN
            <ItemRef Mandatory="No" OrderN
            <ItemRef Mandatory="No" OrderNumb                      ItemOID="LB.LBCAT.HEMATOLOGY"/>
            <ItemRef Mandatory="No" OrderNumber="4" ItemOID="LB.LBCAT.OTHER"/>
            <ItemRef Mandatory="No" OrderNumber="5" ItemOID="LB.LBCAT.URINALYSIS"/>
          </def:ValueListDef>
```

Behind the Scenes of DEFINEExample1.XML

See DEFINEExample1.xml online

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<!-- ********************************************************************************** -->
<!-- File: defineexample1.xml -->
<!-- Date: 28-01-2005 -->
<!-- Author: Clinical Data Interchange Standards Consortium (CDISC) -->
<!-- Description: This is an example define.xml document which implements the Case -->
<!-- Report Tabulation Data Definition Specification Version 1.0.0. -->
<!-- ********************************************************************************** -->
<ODM CreationDateTime="2004-07-28T12:34:13-06:00" FileType="Snapshot" ODMVersion="1.2" FileOID=
xsi:schemaLocation="http://www.cdisc.org/ns/odm/v1.2 define1-0-0.xsd" xmlns:def="http://www.cc
xmlns:xlink="http://www.w3.org/1999/xlink" xmlns:xsi="http://www.w3.org/2001/XMLSchema-ins
  - <Study OID="1234">
    - <GlobalVariables>
        <StudyName>1234</StudyName>
        <StudyDescription>1234 Data Definition</StudyDescription>
        <ProtocolName>1234</ProtocolName>
      </GlobalVariables>
    - <MetaDataVersion OID="CDISC.SDTM.3.1.0" def:StandardVersion="3.1.0" def:StandardName="CDIS
      Data Definitions" Name="Study 1234, Data Definitions">
      - <def:AnnotatedCRF>
          <def:DocumentRef leafID="blankcrf"/>
        </def:AnnotatedCRF>
      - <def:leaf xlink:href="blankcrf.pdf" ID="blankcrf">
          <def:title>Annotated Case Report Form</def:title>
        </def:leaf>
      - <def:SupplementalDoc>
          <def:DocumentRef leafID="SupplementalDataDefinitions"/>
        </def:SupplementalDoc>
      - <def:leaf xlink:href="supplementaldatadefinitions.pdf" ID="SupplementalDataDefinitions">
          <def:title>Supplemental Data Definitions Document</def:title>
        </def:leaf>
```

Reference cdisc.org schema file

Behind the Scenes - Text file of DEFINE.XML

Partial SAS program
to create define.xml

```
filename dheader "define_header.txt";
data define_header;
    set define_header;

    file dheader notitles;

      creationdate = compress(put(datetime(), IS8601DT.));

    put @1 '<?xml version="1.0" encoding="UTF-8" ?>' /
        @1 '<?xml-stylesheet type="text/xsl" href="' stylesheet +(-1) '"?>' /
        @1 '<!--
******************************************************************************** -->' /
        @1 '<!-- File: define2-0-0.xml
-->' /
        @1 "<!-- Date: &sysdate9.
-->" /
        @1 '<!-- Description: Define2-0-0.xml file for '   studyname +(-1) '
-->' /
        @1 '<!--
******************************************************************************** -->' /
        @1 '<ODM' /
        @3 'xmlns="http://www.cdisc.org/ns/odm/v1.3"' /
        @3 'xmlns:def="http://www.cdisc.org/ns/def/v2.0"' /
        @3 'xmlns:xlink="http://www.w3.org/1999/xlink"' /
        @3 'xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"' /
        @3 'xsi:schemaLocation="http://www.cdisc.org/ns/def/v2.0' /
                                '../schema/cdisc.org-define-2.0/define2-0-0.xsd"' /
        %if "&standard" = "ADAM" %then
          @3 'xmlns:adamref="http://www.cdisc.org/ns/ADaMRes/DRAFT"' /
        ;
        @3 'ODMVersion="1.3.2"' /
```

Global Element Order for DEFINE.XML

1. ODM
   a. Study
      i. GlobalVariables
         1. StudyName
         2. StudyDescription
         3. ProtocolName
      ii. MetaDataVersion
         1. def:AnnotatedCRF
            a. def:DocumentRef
         2. def:SupplementalDoc
            a. def:DocumentRef
         3. def:leaf
            a. def:title
         4. def:ComputationMethod
         5. def:ValueListDef
            a. ItemRef
         6. Include
         7. ItemGroupDef ← Domains
            a. ItemRef
            b. def:leaf
               i. def:title
            c. def:ValueListRef
         8. ItemDef ← Variables
            a. CodeListRef
         9. CodeList
            a. CodeListItem
               i. Decode
                  1. TranslatedText
            b. ExternalCodeList

HTML like file that has rules, environment settings, hierarchy with root and child elements, attributes, links, values, lists and internal checks.

See Case Report Tabulation Data Definition Specification DEFINE.XML reference.

Global Element Order for DEFINE.XML

Dataset Definitions

- **ItemGroupDef** element, Include references to variable definitions, Include references to actual data file
- **Variable references:** ItemDef element
- **Dataset definition**: def:leaf element

Attributes

- **Repeating**: 'No' for datasets with one record per subject 'Yes' for datasets with more than one record per subject
- **Mandatory**: 'Yes' for required variables, 'No' for Expected or Permissible variables
- **Domain**: Name of the domain
- **IsReferenceData**: 'Yes' for reference datasets (no subjects) 'No' for subject-level datasets

Global Element Order for DEFINE.XML

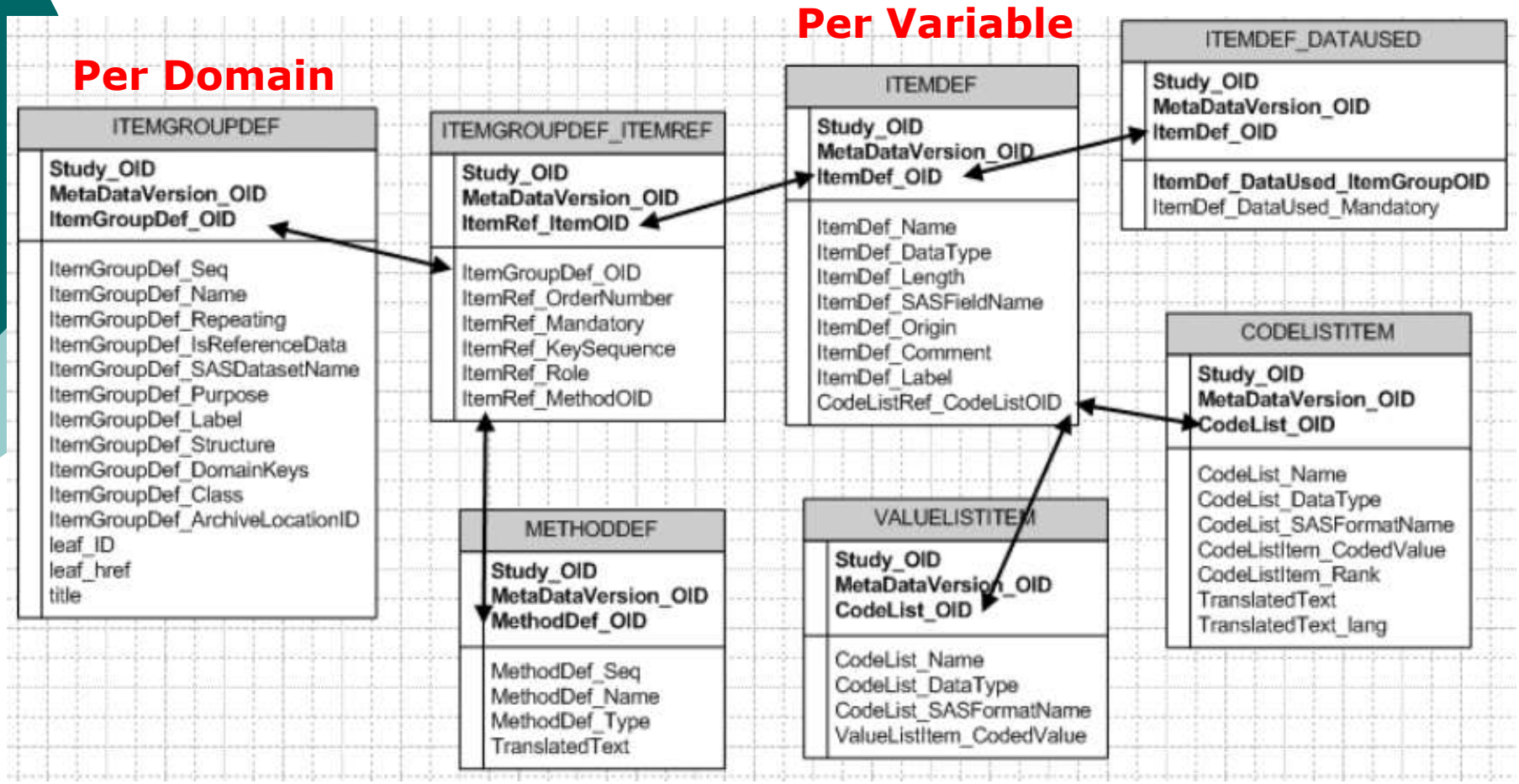
Variable Definitions

Variable definitions
- ItemDef element

Other elements
- Controlled terminology: CodeList
- Value level metadata: def:ValueListDef
- Computational method: MethodDef
- Comments: def:CommentDef
- Origin: def:Origin

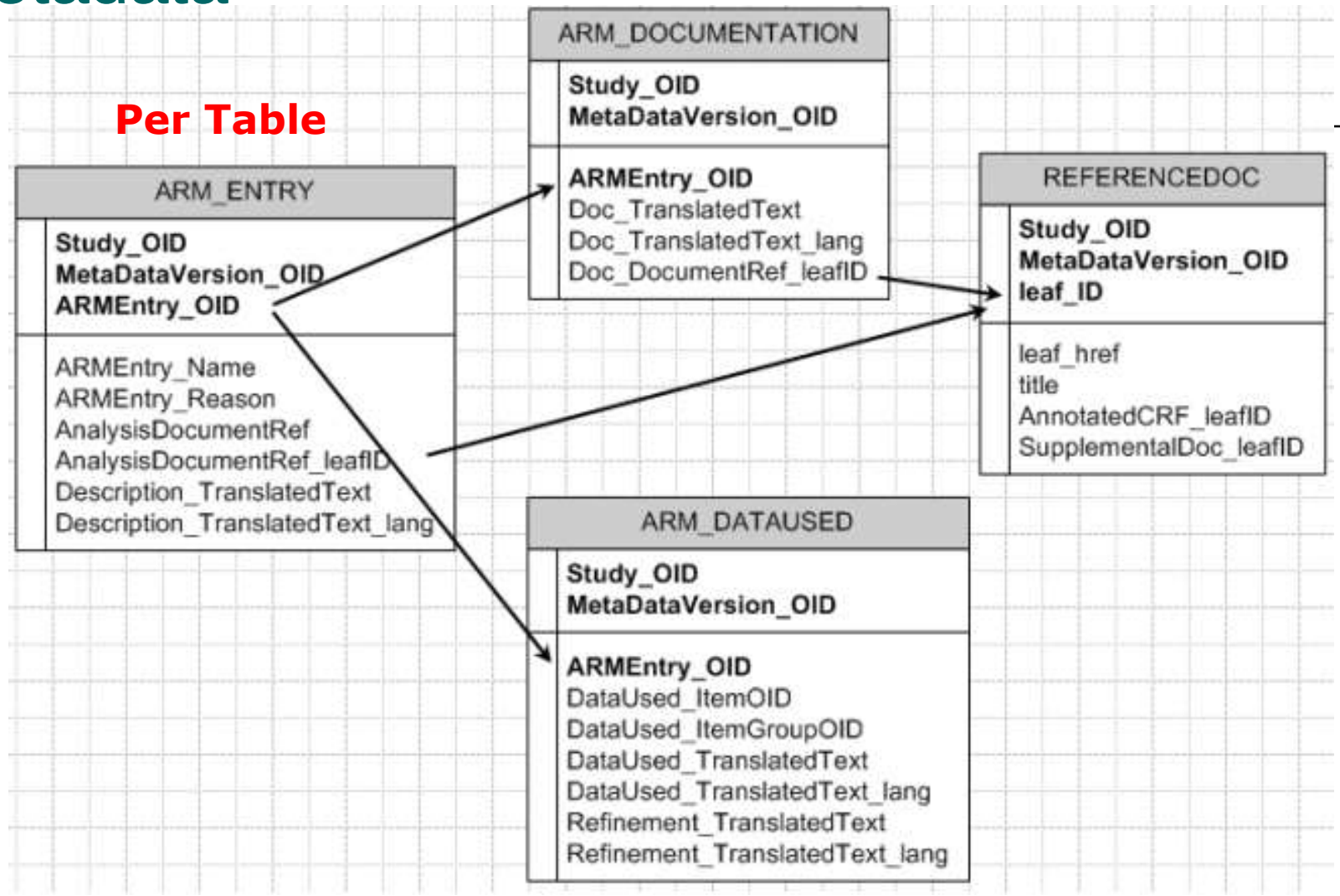
Data types
- Text ($), Integer (#), Float (continuous #)
- Datetime, Date, Time
- PartialDate, PartialTime, IncompleteDatetime
- DurationDatetime

Relational Model for DEFINE.XML



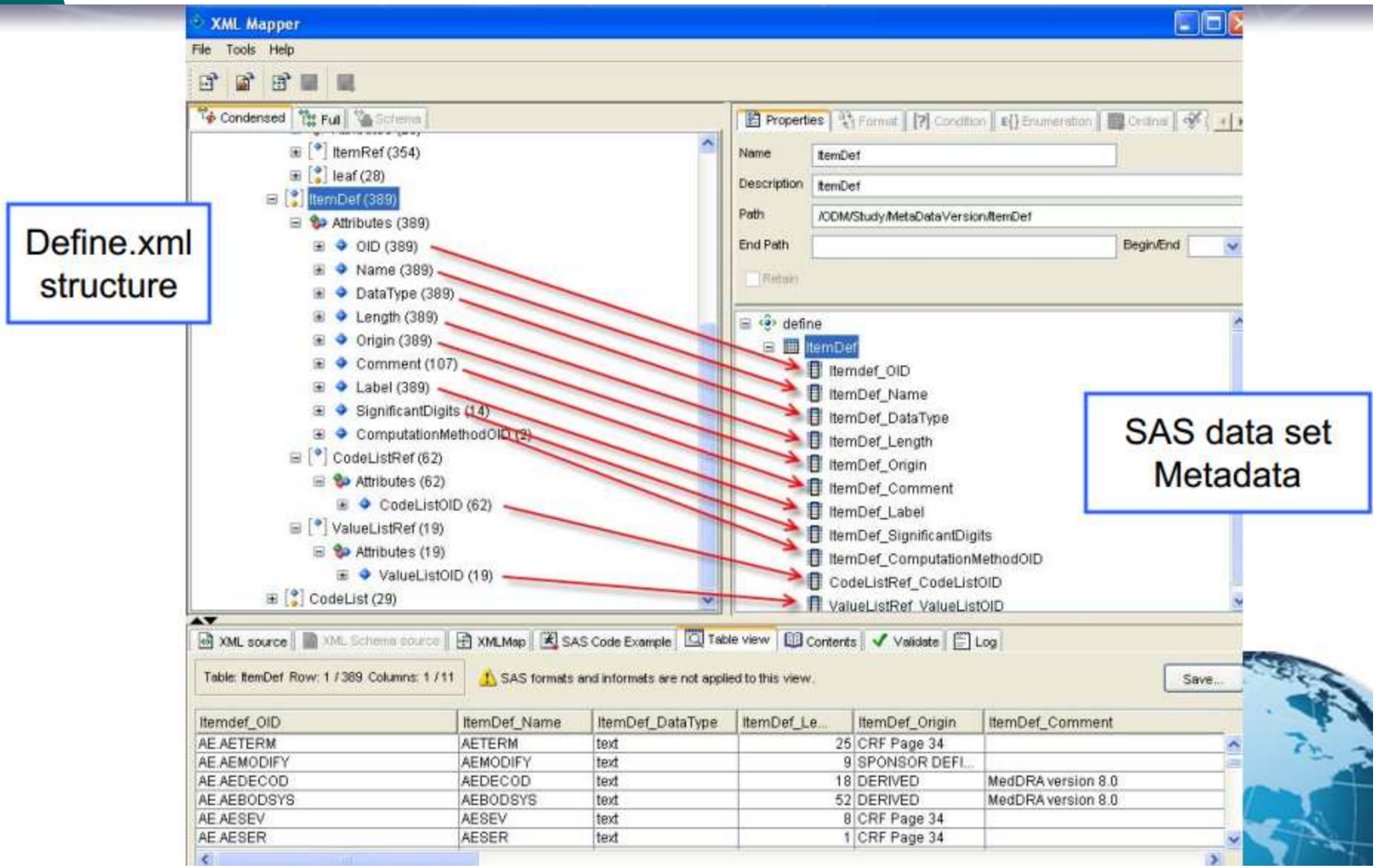
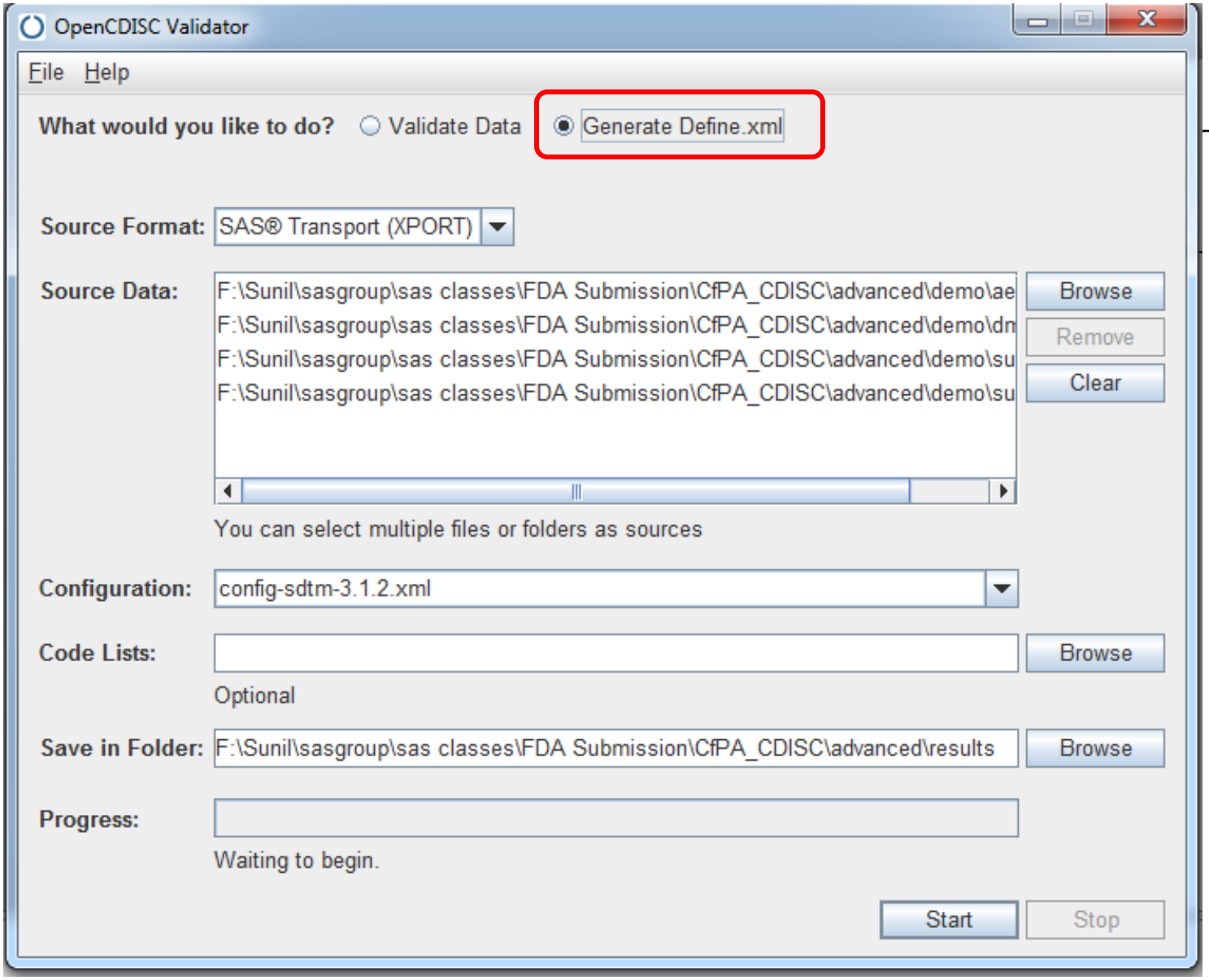
All files are linked by key variables: XXX_OID. Content is grouped by how it will constructed in DEFINE.XML.

Relational Model for Analysis Results Metadata



All files are linked by key variables: XXX_OID. Content is grouped by how it will constructed in DEFINE.XML.

Use SAS XML Mapper to view more complicated DEFINE.XML file and create meta SAS Datasets



OpenCDISC to create DEFINE.XML